

Es. Dimensionamento fl. H₂ A 6V

Dati:

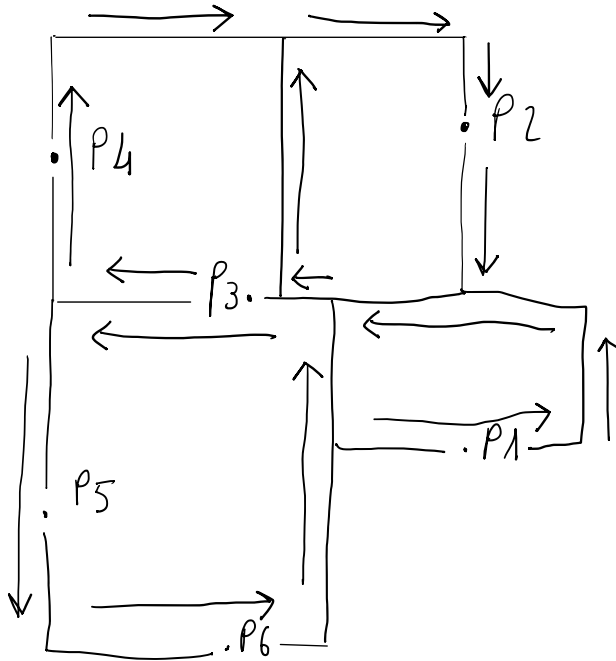
- $V_c = 0.5 \text{ m/sec}$. $V_{sc} = 0.64 \text{ m/sec}$. $T_c = 20 \text{ sec}$
- $T_{sc} = 10 \text{ sec}$. Duratione turno = 7.5 h
- From / to chart (uel turno)

	P1	P2	P3	P4	P5	P6
P1		200	160	200	100	80
P2	20				200	
P3		80			240	
P4						120
P5		160				
P6			10			

• Posizioni dei punti

⇒ Determinare il reticolo
(versi di percorrenza)

⇓
Calcolare le distanze



d_{ij} (m)	P1	P2	P3	P4	P5	P6
P1		42	22	30	34	44
P2	54				28	
P3		32			12	
P4						62
P5		52				
P6			22			

(From/to
chart delle
distanze dei
viaggi carrelli
carichi)

Determinare i tempi di missione $\left(\frac{\text{distanza}}{V_c} + \overline{t_c} + \overline{t_{sc}} \right)$

t''_{ij}	P1	P2	P3	P4	P5	P6	
P1		114	74	90	98	118	$\frac{42}{0.5} = 84 \text{ sec}$
P2	138				86		$84 + 10 + 20 = 114 \text{ sec}$
P3		94			54		
P4						154	
P5		134					
P6			74				

$$H' = \sum_j \sum_i \left([t''_{ij}] \times [n_{ij}] \right) = H' \quad \left(\begin{array}{l} \text{tempo totale} \\ \text{e turno per} \\ \text{tutte le missioni} \end{array} \right)$$

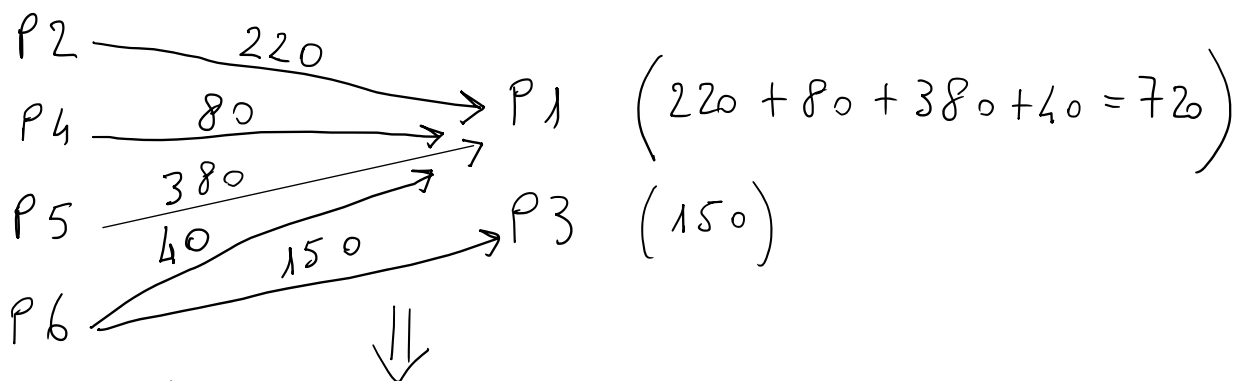
$H' = 152.980 \text{ [sec]}$

Bilanciamento (carrelli vuoti)

m_{ij}	P1	P2	P3	P4	P5	P6	da	NF
P1							740	-720
P2							220	220
P3							320	-150
P4							120	80
P5							160	380
P6							10	190
Σ	20	440	170	200	540	200		



Soluzione di bilanciamento



⇓

x'_{ij}	P1	P2	P3	P4	P5	P6
P1						
P2	220					
P3						
P4	80					
P5	380					
P6	40		150			

$d_{is} [m]$ P_1 P_2 P_3 P_4 P_5 P_6

P_1

P_2 54

P_3

P_4 78

P_5 26

P_6 16

22



t_{is}

P_1 -

P_2 (84)

P_3

P_4 122

P_5 41

P_6 25

34

$$\frac{54}{0.64} = 84 \text{ sec}$$

$$\frac{[m]}{[m/sec]} = [sec]$$

$$\sum_i \sum_j [t'_{ij}] \times [n'_{ij}] = \Delta H = 49906 \text{ [sec]}$$

$\downarrow$ temp. minime di lancio
 $\downarrow$ # veggi di lancio

$$H = H' + \Delta H = 152980 + 49906 = 202.886 \text{ [sec]}$$

$$= 56.4 \text{ [ore]}$$

$$\Rightarrow \# \text{ carrelli} = \frac{H}{7.5} = \frac{56.4}{7.5} = 7.51$$

$$\Rightarrow 8 \text{ carrelli con un tasso di utilizzazione teorico } u = \frac{7.51}{8} = 0.94 \text{ (94\%)}$$